Explainable Rule-Based Clinical Decision Support System for Breast Cancer Using NLP

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***Abstract*—**Clinical decision making in the case of breast cancer patients is done taking into account various clinical parameters like Estrogen Receptor (ER), Progesterone Receptor (PR), Human Epidermal Growth Factor Receptor 2 (HER2), Ki-67 index, tumor size, lymph node involvement, and disease stage. All these parameters are scattered throughout the patient's reports, which makes the task of extracting all such relevant data time-consuming. CDSSs would be helpful in this process, as they would help to organize the information and make clinical decisions accordingly. Unfortunately, many AI-based CDSS systems that exist today mostly rely on the prediction of the disease and give recommendations without any explanation of their choices.

The following paper presents an explainable rule-based CDSS that uses NLP to analyze the reports and provide recommendations for the treatment of breast cancer patients. Our system analyses the reports and extracts relevant entities in order to form a patient's profile. Further, the biomarkers' values are analyzed in order to determine the possible rule-based clinical subtype and treatment category of breast cancer that include Luminal A-like, Luminal B-like, HER2-positive, and TNBC. The guideline-based decision engine applies formalized treatment rules based on well-known breast cancer treatment guidelines in order to create treatment recommendations. Moreover, the safety layer will take into account patient comorbidities, contraindications, and other risks prior to making the recommendation.

The system implements the clinician-in-the-loop paradigm, in which the doctor is the ultimate decision-maker and he can confirm, edit, and/or reject the proposed treatment algorithm. The proposed architecture takes into account not only clinician input but also treatment outcomes in order to improve the recommendation process in the future. Experiments related to information extraction through NLP were implemented in order to explore the feasibility of such a solution.

The experiments are shown as an initial implementation study, while the entire system is supposed to be an explainable and clinically supervised recommendation system.

***Index Terms*—Breast Cancer, Clinical Decision Support System, Natural Language Processing, Explainable AI, Rule-Based System, ClinicalBERT, Treatment Recommendation, Clinician-in-the-Loop**

I. Introduction

Breast cancer is amongst the most frequently diagnosed forms of cancer and it needs a thorough analysis before deciding on an effective treatment plan. There are certain clinical and pathological features that need to be considered including Estrogen Receptor (ER), Progesterone Receptor (PR), Human Epidermal Growth Factor Receptor 2 (HER2), Ki-67 Index, tumor size, lymph node involvement, and disease stage. The aforementioned details may be found in various pathology reports, biopsy reports, and clinical notes. It is essential for the patient's data collection and interpretation.

In terms of molecular subtyping and clinical biomarker usage, breast cancer can be classified in different ways. Amongst the main molecular subtypes one can mention Luminal A, Luminal B, HER2-enriched, and basal-like breast cancer [1]. In addition, clinical biomarkers, such as ER, PR, HER2, and Ki-67 are used in the treatment-oriented classification which involves hormone receptor-positive, HER2-positive, and triple-negative breast cancer. Each type has a unique treatment approach. For example, hormone receptor-positive breast cancer needs endocrine therapy, HER2-positive breast cancer requires HER2 targeted therapy, and TNBC usually goes with chemotherapy. Other clinical considerations like tumor size, lymph nodes involvement, and disease staging are also significant when deciding on the right treatment pathway.

Clinical Decision Support System (CDSS) has been examined as one of the ways that could assist in making decisions about cancer treatment based on patient data and clinical

expertise [2], [3]. Many AI-based systems in recent times have concentrated on tasks like detection, prediction, prognosis, or even treatment reaction for cancers [4]–[6]. However, many of these AI systems operate as black boxes, which means it may be challenging for the physician to understand the rationale behind the prediction or recommendation made [7], [8].

Natural Language Processing (NLP) may assist in converting clinical unstructured text to structured information. In recent research, the use of domain-specific language models and NLP techniques have been considered for identifying the phenotypes and pathology of breast cancer from clinical reports [9], [10]. Such methods prove the feasibility of automatic identification of important clinical information like receptor status, tumor properties, and staging. However, information extraction is not sufficient for a complete treatment decision process.

In this paper, we introduce an explainable rule-based Clinical Decision Support System for treatment of breast cancer through NLP. The system extracts relevant clinical information from the diagnostic reports and generates a structured patient profile. The information on the extracted biomarkers helps to determine the probable molecular subtype of the patient. Then a rule-based decision engine makes treatment decisions by applying formalized treatment rules based on existing treatment guidelines on breast cancer like NCCN, ESMO, ASCO, and Indian treatment guidelines if available [11]–[16].

The recommended system is also accompanied by a safety module that takes into account the presence of any contraindications, comorbidities, allergic reactions, and others. This recommendation is not meant to substitute for the physician. Rather, a physician-in-the-loop method is employed, where the physician may either accept, alter, or deny the suggested treatment process. Additionally, the reasoning behind the recommendation is also given, so that the physician understands the underlying clinical facts and rules.

The contributions of this research include the following:

- An NLP-based pipeline for extracting important breast cancer biomarkers and clinical entities from diagnostic reports.
- A structured approach for molecular subtype classification using ER, PR, HER2, and Ki-67 information.
- An explainable rule-based treatment recommendation engine based on established breast cancer treatment guidelines.
- A safety layer that considers comorbidities, contraindications, allergies, and clinical risks before presenting treatment options.
- A clinician-in-the-loop feedback mechanism where doctors can approve, modify, or reject recommendations for future improvement.

The rest of this paper is organized as follows. Section II discusses related work and the research gap. Section III presents the proposed system design and methodology. Section IV describes the implementation and experimental study. Section V discusses the results and analysis. Finally, Section VI presents the conclusion and future work.

II. Related Work and Research Gap

A. Clinical Decision Support Systems for Breast Cancer

Clinical Decision Support Systems (CDSS) have been investigated as a means of assisting healthcare professionals in diagnosis, treatment planning, and clinical decision-making. These systems combine patient-specific information with medical knowledge, clinical guidelines, and decision logic to support the selection of appropriate clinical pathways. Mazo *et al.* discussed the application of CDSS in breast cancer and highlighted their potential to support clinical decisions using patient data and clinical knowledge [2]. Similarly, Andrzejewski *et al.* explored the use of formalized clinical guidelines and expert decision patterns to support breast cancer-related decisions [3].

B. Comparison with Existing Research

Table I summarizes the major research themes reviewed above and identifies how the proposed Oncopilot workflow builds on them. The comparison describes the intended scope of the proposed system; it does not imply that every feature is absent from all previously published systems.

These studies establish the importance of representing clinical knowledge in a structured form so that it can be applied systematically to individual patient cases. Guideline-based reasoning can improve consistency and make decision logic easier to inspect. However, a complete clinical workflow also requires reliable extraction of information from medical documents, verification of the extracted values, assessment of missing information, and presentation of the resulting pathway to the clinician. These requirements motivate the integration of document processing, clinical information extraction, decision logic, and clinician review within a unified system.

C. Artificial Intelligence in Breast Cancer Diagnosis and Treatment Planning

Artificial Intelligence (AI), including machine learning and deep learning, has been investigated for breast cancer diagnosis, prognosis, treatment response prediction, and personalized clinical support. These approaches can identify patterns in large medical datasets and support the analysis of complex relationships between clinical variables and disease outcomes [4]–[6]. In breast imaging, machine learning and deep learning methods have also been widely studied for cancer detection and classification [7].

Although predictive models can provide useful information, their outputs do not necessarily explain the clinical evidence or reasoning underlying an individual prediction. The interpretability of AI systems is therefore important when their outputs are intended to support high-stakes clinical decisions. Ghasemi *et al.* discussed issues related to explainability and trust in AI-based healthcare applications [8].

For breast cancer treatment planning, a prediction alone may not be sufficient. Clinicians also need to understand which patient characteristics influenced the result, which clinical rules were applied, and whether important information is missing. Oncopilot therefore emphasizes traceable, guideline-based

TABLE I
COMPARISON OF RELATED RESEARCH AREAS AND THE PROPOSED ONCOPILOT FRAMEWORK

Research Area	**Representative Studies**	**Contribution and Scope of Existing Work**	**Proposed Oncopilot Approach**
Clinical decision support	[2], [3]	Discusses CDSS applications, clinical knowledge representation, formalized guidelines, and structured decision support for breast cancer.	Integrates document processing, structured clinical information, guideline reasoning, safety checks, and clinician review within one workflow.
AI for breast cancer	[4]–[7]	Investigates AI for clinical prediction, cancer analysis, diagnosis, prognosis, or image-based detection and classification.	Uses NLP to structure report-derived clinical facts and combines them with explicit guideline-based reasoning rather than relying solely on a predictive output.
Clinical NLP and information extraction	[9], [10]	Provides domain-specific language modelling and clinical text-processing approaches relevant to identifying medical entities and contextual information.	Targets extraction of ER, PR, HER2, Ki-67, tumor characteristics, lymph-node findings, and staging information, followed by normalization and validation.
Biomarker-based stratification	[1], [9]	Establishes the relevance of breast cancer biological characteristics and cancer-related phenotype extraction.	Uses available biomarkers together with tumor characteristics and disease extent to support treatment-oriented clinical stratification.
Guideline-based reasoning	[3], [11]–[14]	Provides clinical knowledge and evidence-based guidance that can inform structured treatment decision logic.	Represents clinically verified guidance as versioned rules with explicit conditions, pathway references, and traceable decision rationale.
Explainability and trust	[2], [8]	Discusses clinical decision support and the importance of interpreting AI-supported healthcare outputs.	Links each proposed pathway to extracted patient facts, matched rules, guideline sources, and identified uncertainties.
Safety and contraindications	[2], [6], [8]	Provides background on clinical decision support and AI-assisted healthcare, motivating attention to patient-specific safety and clinical oversight.	Adds a dedicated safety-checking stage for available comorbidity, allergy, organ-function, medication, and contraindication information.
Clinician-in-the-loop review	[2], [6], [8]	Supports the need for clinical oversight and appropriate human involvement in AI-assisted decisions.	Allows clinicians to verify extracted information and accept, modify, or reject the proposed clinical pathway.
Feedback and evaluation	[4], [6], [8]	Provides related context for AI-supported clinical systems and motivates systematic performance assessment.	Records reviewed decisions and evaluation errors to support controlled improvement, with validation before any model or rule changes.
Integrated workflow	**Combined research themes above**	**The cited research areas address complementary parts of clinical information processing and decision support. A system-level comparison is required to establish the extent of integration in individual published systems.**	**Connects report processing, clinical extraction, validation, stratification, guideline reasoning, safety, explanation, and clinician review in one proposed framework. Its integrated performance will be evaluated experimentally.**

reasoning rather than treating a model-generated prediction as an independent treatment decision.

D. Natural Language Processing for Clinical Information Extraction

Pathology reports and other clinical documents contain important information about tumor characteristics, biomarkers, histology, and disease extent. However, these details may be expressed through different terms, abbreviations, report sections, and narrative formats. Extracting this information manually can require considerable effort and may complicate the preparation of structured patient profiles.

Natural Language Processing (NLP) provides techniques for converting unstructured clinical text into structured information. Zhou *et al.* introduced CancerBERT for identifying cancer-related phenotypes in clinical text, including breast cancer-related information [9]. ClinicalBERT research has also demonstrated the relevance of domain-adapted language representations for processing clinical notes [10].

These studies support the use of domain-specific language models and information extraction techniques in clinical document processing. Named Entity Recognition (NER), contextual language representations, and rule-based pattern matching can be used to identify relevant clinical entities and their associated values. For Oncopilot, the target information includes estrogen receptor (ER), progesterone receptor (PR), human epidermal growth factor receptor 2 (HER2), Ki-67, tumor size, histological type, tumor grade, lymph-node status, and TNM or disease stage where available.

However, entity identification alone does not produce a complete clinical profile. The system must also determine whether a finding is positive, negative, uncertain, absent, or contradictory. For example, a missing HER2 result must not be interpreted as a negative result. Similarly, a mention of breast cancer in a family history section must not be treated as a confirmed finding for the patient. These requirements motivate the addition of contextual validation, normalization, and missing-information handling after information extraction.

E. Biomarker-Based Clinical Stratification

Biological and molecular characterization is an important component of breast cancer assessment. Perou *et al.* described molecular portraits of human breast tumours and demonstrated the biological heterogeneity of breast cancer [1]. In clinical practice, biomarkers such as ER, PR, HER2, and Ki-67, together with histopathological features and disease extent, provide important information for treatment planning.

These variables must be interpreted in combination rather than as isolated indicators. Tumor size, lymph-node involvement, histological grade, TNM classification, disease stage, and relevant patient factors can affect the clinical context in which a treatment pathway is considered.

Oncopilot uses extracted biomarker information as an input to treatment-oriented clinical stratification. This stage organizes the available findings into a structured clinical category for subsequent guideline-based reasoning. It does not assume that immunohistochemical findings alone establish a definitive gene-expression-based molecular subtype. Where the required information is incomplete or inconsistent, the system should preserve the uncertainty and request clinical verification rather than infer unsupported values.

F. Guideline-Based Clinical Reasoning

Clinical guidelines provide evidence-based recommendations for breast cancer assessment and treatment. Guidance from organizations such as the National Comprehensive Cancer Network (NCCN), European Society for Medical Oncology (ESMO), and American Society of Clinical Oncology (ASCO), along with relevant Indian guidance, can inform the development of structured clinical decision logic [11]–[14].

A computational rule engine can represent verified recommendations as explicit conditions and corresponding clinical pathways. These conditions may refer to biomarker status, disease stage, tumor characteristics, lymph-node involvement, and other relevant patient information. The engine can then evaluate a structured patient profile against the applicable rules and record which conditions were satisfied, which were not satisfied, and which could not be evaluated because of missing information.

Compared with an approach that generates recommendations directly from free-text input, explicit rule-based reasoning offers a more inspectable and reproducible decision process. It also allows each rule to be associated with its source, version, and clinical rationale. Nevertheless, rule-based systems require careful maintenance because clinical guidelines can change. The rules must therefore be reviewed, clinically verified, version-controlled, and updated when the underlying guidance changes. The system must also avoid selecting a pathway when essential eligibility information is unavailable.

G. Explainability, Safety, and Clinician Involvement

Explainability and clinical oversight are important considerations in the development of healthcare decision support systems. Existing work on CDSS and AI-supported healthcare highlights the importance of supporting clinical professionals rather than replacing their judgement [2], [6], [8].

A treatment pathway may be affected by factors beyond biomarker status and tumor characteristics. Comorbidities, allergies, organ function, concomitant medications, and other contraindications may require additional evaluation. Consequently, a treatment-support system should distinguish between a pathway identified by the guideline engine and a pathway that has been fully assessed for the individual patient.

Oncopilot incorporates a safety-checking stage to identify available contraindication information, missing safety data, and issues requiring further clinical review. Its explanation layer is intended to connect the extracted clinical findings with the applicable decision rules and their supporting guideline references. This enables the clinician to inspect the basis of the proposed pathway instead of receiving an unexplained output.

The clinician remains responsible for the final decision. The proposed workflow allows the clinician to review the extracted profile, verify uncertain findings, and accept, modify, or reject the suggested pathway. Feedback can be recorded for system evaluation and controlled future improvement. Such feedback must be reviewed and validated before it is used to modify clinical rules or retrain the NLP model.

H. Research Gap and Motivation for Oncopilot

The reviewed literature addresses several important components of breast cancer decision support. CDSS research provides approaches for representing clinical knowledge and supporting decisions. AI research contributes methods for analyzing medical data and predicting clinical outcomes. Clinical NLP enables the extraction of information from unstructured documents, while biomarker research supports the characterization of breast cancer. Clinical guidelines provide a basis for structured treatment reasoning.

However, these research directions address different parts of the overall clinical workflow. For Oncopilot, the key research problem is how to integrate these components into a coherent and traceable process that starts with unstructured pathology reports and ends with a clinician-reviewed clinical pathway.

The resulting research gap can be expressed through the following requirements:

- **Integrated information extraction:** Clinical reports must be transformed into a structured profile containing treatment-relevant information.
- **Context-aware validation:** Extracted findings must be normalized and checked for negation, uncertainty, missing values, and conflicting information.
- **Guideline-grounded reasoning:** Structured patient information must be evaluated using explicit, clinically verified, and version-controlled decision rules.
- **Safety-aware pathway generation:** Relevant contraindications, patient factors, and incomplete safety information must be identified before a pathway is presented for review.

- **Traceable explanations:** The system must connect its proposed pathway to the patient facts, applicable rules, and supporting guideline sources.
- **Clinician oversight and evaluation:** Clinicians must be able to verify information and review the proposed pathway, while system performance is evaluated using appropriate quantitative and clinical measures.

The proposed Oncopilot framework addresses these requirements through a connected workflow comprising clinical document processing, NLP-based information extraction, clinical data validation, treatment-oriented stratification, guideline-based decision logic, safety checks, explainability, and clinician review. Its intended contribution is the integration of these components into an auditable clinical decision-support workflow, rather than the introduction of a new cancer treatment guideline or an autonomous treatment-prediction model.

The effectiveness of this integration must be established through experiments measuring extraction performance, handling of missing or conflicting information, guideline concordance, safety-check performance, and the faithfulness of the generated explanations.

III. Proposed System and Methodology

The system that will be built is the Explainable Clinical Decision Support System intended to help doctors when they plan breast cancer treatment. It will not substitute the doctor or make any clinical decisions by itself; on the contrary, it will gather the necessary information from clinical documents, structure it for a certain patient, evaluate the potential treatment strategies according to certain guidelines, and provide the explanations for each suggestion.

The entire process includes the following steps: document intake, clinical information extraction, data validation, molecular subtyping, treatment reasoning based on guidelines, safety check, recommendations generation, and feedback from the clinician.

A. Document Ingestion and Preprocessing

The system accepts clinical information in the form of pathology reports, biopsy reports, PDFs, text files, and Electronic Health Records (EHR). These documents may contain both structured and unstructured clinical information.

The input text goes through pre-processing in this step. This involves parsing the document and converting the content to machine-readable text. OCR may be applied if there is any scanned text in the document. Formatting elements and unnecessary elements are stripped out, whereas common medical abbreviations and expressions are standardized. The preprocessed text is then tokenized and segmented into sentences.

This process gets the clinical text ready for entity extraction.

B. NLP-Based Clinical Information Extraction

The task of entity recognition is carried out using the NLP module, which pulls out relevant clinical entities related to breast cancer from the report. This process involves both pattern matching using rules and spaCy based Natural Language Processing. The concept of ClinicalBERT has also been studied experimentally.

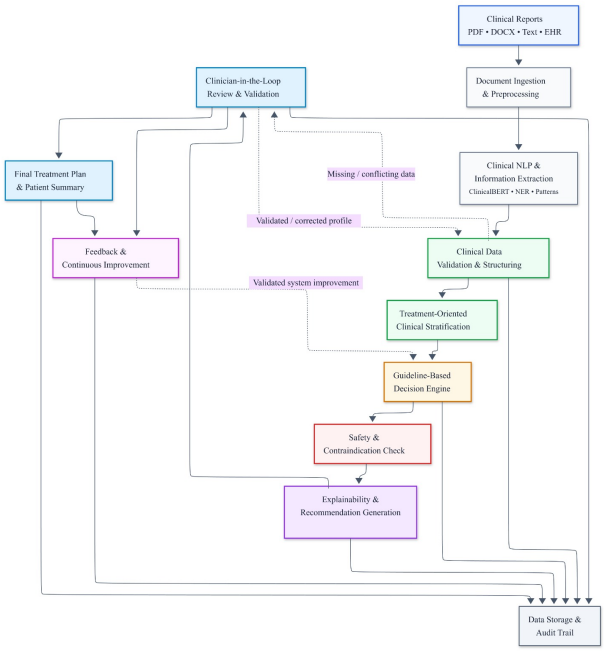


Fig. 1. Overall architecture of the proposed Oncopilot clinical decision support system, illustrating clinical information extraction, data validation, treatment-oriented stratification, guideline-based reasoning, safety checks, clinician review, feedback, and audit logging.

The important entities considered by the system include:

- Estrogen Receptor (ER) status
- Progesterone Receptor (PR) status
- Human Epidermal Growth Factor Receptor 2 (HER2) status
- Ki-67 proliferation index
- Tumor size
- Lymph node involvement
- TNM stage
- Disease stage
- Comorbidities and relevant clinical conditions
- Allergies and treatment-related contraindications

Rule-based pattern helps detect the commonly mentioned values like positive, negative, percent value, tumor size, and lymph node status. The use of ClinicalBERT has been incorporated in order to make sure that clinical language is taken into consideration while dealing with the context which may not be following any structured report format.

This extracted data is then used to generate the structured patient profile.

C. Clinical Data Validation

The patient profile obtained from the database is examined for any missing data, partial data or unclear data. Receptor status, tumor characteristics and disease stage among other

important fields are examined and validated before passing them to the decision engine.

In the event that a value is missing or not clearly defined, then the system alerts the clinician to verify the missing value instead of assuming it. This minimizes the risk of recommending based on partial clinical data.

D. Molecular Subtype Classification

The system uses the extracted biomarker information to assign a rule-based clinical subtype or treatment-oriented category.The classification uses ER, PR, HER2, and Ki-67 values to determine the possible molecular characteristics of the tumor.

The rule-based classification considers the following general patterns:

- ER-positive, PR-positive, HER2-negative, and low Ki-67 may be classified as Luminal A.
- Hormone receptor-positive tumors with higher Ki-67 or other characteristics associated with increased proliferation may be classified as Luminal B.
- HER2-positive status is used to identify cases that may require evaluation for HER2-targeted treatment pathways.
- ER-negative, PR-negative, and HER2-negative tumors may be classified as Triple-Negative Breast Cancer (TNBC).

The identified subtype is used as one of the main inputs for treatment pathway generation. Tumor size, lymph node involvement, and disease stage are also considered to provide a broader clinical context.

E. Guideline-Based Decision Engine

The key element of the new system is a transparent decision engine based on rules. The knowledge from the standard treatment protocols of breast cancer is transformed into IF–THEN rules.

The rules assess the structured profile of the patient that includes the molecular type of cancer, receptor status, tumor features, lymph node status, cancer staging, and other clinical data. From the selected rules, the system provides possible treatment options.

Depending on the patient profile, these pathways may include:

- Endocrine therapy
- HER2-targeted therapy
- Chemotherapy
- Combination treatment approaches
- Immunotherapy-related treatment pathways where clinically applicable

Each recommendation is linked to the clinical factors and decision rules that led to its selection. This makes the reasoning process visible to the clinician.

F. Contraindication and Safety Checking

Before making any recommendations for treatment, the system assesses relevant safety factors. The safety module

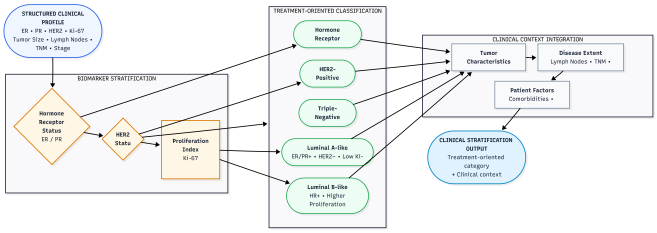


Fig. 2. Biomarker-driven clinical stratification for breast cancer treatment planning. The structured clinical profile integrates ER, PR, HER2, and Ki-67 with tumor characteristics, lymph-node involvement, TNM stage, disease stage, and relevant patient factors to derive a treatment-oriented clinical category for downstream guideline-based decision making.

evaluates any information available with regard to the comorbidities, allergies, organ functioning, and any contraindications for the particular treatment.

This safety module is used to evaluate those instances whereby the suggested treatment may need further clinical evaluation. Instead of excluding the clinician from the decision-making process, the system draws attention to the relevant safety issue and provides it alongside the treatment suggestion.

This component is added in order to make the decision support more patient-specific and avoid suggesting a treatment that does not take into account certain clinical factors.

G. Recommendation Ranking and Explanation

The system evaluates the applicable treatment pathways after clinical data extraction, subtype classification, guideline-based reasoning, and safety checking. The recommendations can then be ranked based on their relevance to the available patient information and the applicable clinical rules.

The output presented to the clinician contains the following:

- Extracted clinical profile
- Identified molecular subtype
- Possible treatment pathway
- Applicable guideline-based rule
- Clinical factors used in the recommendation
- Safety flags, contraindications, or missing information

The focus of this output is explainability. The clinician should be able to understand why a particular treatment pathway was suggested instead of receiving only a final prediction.

H. Clinician-in-the-Loop Feedback

The proposed system keeps the doctor as the final decision-maker. The clinician can approve the suggested treatment pathway, modify it based on additional clinical knowledge, or reject it.

The clinician's decision and relevant treatment outcomes can be recorded as feedback. This information can later be used to study the performance of the system and improve the recommendation process.

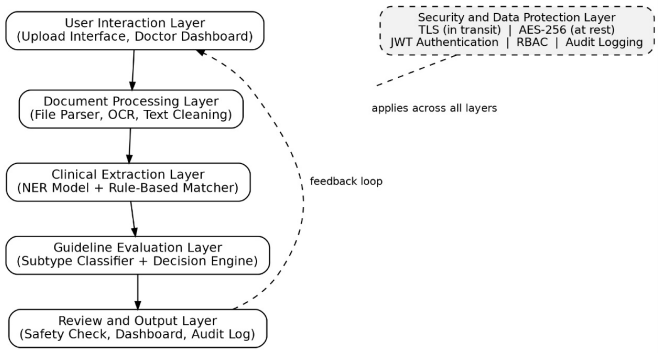


Fig. 3. Layered architecture of the proposed clinical decision support system.

The feedback mechanism supports future development of adaptive components without removing the explainable rule-based foundation of the system. Any future learning-based improvement is intended to support the clinical decision process while maintaining clinician supervision and transparent treatment reasoning.

IV. Implementation and Experimental Study

A. System Implementation

The suggested system is implemented as a modular CDSS, where each stage completes a different task of the treatment support process. The implementation process starts from ingesting and processing the clinical documents. The diagnostic reports may be submitted in the form of text or document file and then the corresponding clinical content is preprocessed to be analyzed.

The NLP module is implemented based on the Python-related technologies. The identification of structured clinical information from the text is conducted using spaCy and rule-based pattern matching. During the first phase of the implementation, ClinicalBERT model was also studied for the exploration of the usage of the clinical language model in understanding and analyzing medical text.

The identified information is structured in the patient profile. The profile includes all the necessary information about ER status, PR status, HER2 status, Ki-67 index, tumor size, lymph nodes status, TNM information, disease stage, and other relevant clinical aspects.

The architecture for the proposed application consists of a backend, which uses Python, and a frontend based on the web. The backend is responsible for performing tasks such as clinical data processing, entity extraction, validation, rules evaluation, and recommendation generation. The frontend would be used to present the clinicians with the extracted patient profile and recommendations along with other details.

B. Experimental NLP Pipeline

A preliminary pipeline in NLP was created on Google Colab in order to evaluate the potential of mining the breast cancer information in the clinical texts. It was designed as an early experimental implementation of the information extraction module in the proposed solution.

The following steps were involved: document and text preprocessing, text cleaning, sentence segmentation, spaCy processing, rule-based entity extraction, and ClinicalBERT experiments. During the information extraction process, emphasis was placed on the extraction of the clinically significant information: receptor status, values of Ki-67, tumor features, information about lymph nodes, etc.

Rule-based patterns were especially helpful when extracting the structured expressions typical of clinical reports. Such data includes positive/negative receptor status, percent values of Ki-67, tumor measurements, and information about lymph nodes. The extracted information would be used in the structured fields by the molecular subtype classification and decision engine.

ClinicalBERT was explored to understand whether a clinical domain-specific language model could improve the interpretation of medical text. The experimental results showed that language model-based processing can provide useful clinical context. However, the initial implementation also showed that model performance depends heavily on the quality, format, and quantity of the available clinical text.

C. Rule Extraction and Decision Engine Prototype

In addition, the potential of a rule extraction algorithm from the guideline and clinical text was examined. The aim of this experiment was to test the possibility of formulating relevant conditions for treatment in a structured way and verifying them with respect to patient information.

The created rules included conditions depending on various clinical features like HER2 status and presence of lymph nodes. Every rule had its own outcome, confidence and description. According to the experimental data, automatically produced rules may include inaccurate and misinterpreted statements from the clinical text due to its noise and poor structure.

Such a conclusion became one of the key reasons for choosing the clinically validated rules as an input for the proposed decision support engine instead of the automatically extracted ones. In the final system, treatment rules should be created with respect to the existing clinical guidelines.

D. Evaluation Approach

The NLP extraction component can be evaluated using standard information extraction metrics. The extracted clinical entities can be compared with manually verified reference annotations.

Precision measures the proportion of extracted entities that are correct:

$$Precision = \frac{TP}{TP + FP}$$

where TP represents correctly extracted entities and FP represents incorrectly extracted entities.

Recall measures the proportion of relevant clinical entities that are successfully identified:

$$Recall = \frac{TP}{TP + FN}$$

where FN represents relevant entities that were not extracted by the system.

The F1-score provides a combined measure of precision and recall:

$$F1 = 2 \times \frac{Precision \times Recall}{Precision + Recall}$$

These metrics can be calculated separately for different clinical entities such as ER, PR, HER2, Ki-67, tumor size, lymph node involvement, and disease stage.

E. Test Case Design

The proposed system is designed to be tested using representative breast cancer case scenarios. The test cases cover different combinations of biomarker status, tumor characteristics, and disease stages.

The cases include profiles representing the major molecular subtypes:

- Luminal A
- Luminal B
- HER2-positive or HER2-enriched treatment pathway
- Triple-Negative Breast Cancer

Variation may also be added based on tumor size, involvement of lymph nodes, disease staging, comorbidities, allergies, and contraindications. This way, the rule engine and safety module can be assessed in various medical scenarios.

The assessment of the proposed system as a whole will be performed taking into account not only the precision of information extraction but also the correctness and explainability of the pathway generation. The generated recommendations can be compared with corresponding clinical rules and assessed by domain specialists.

V. Preliminary Experimental Findings and Discussion

A. NLP Extraction Results

The initial implementation was used for testing the feasibility of obtaining breast cancer-related information from report text. The pipeline included text preprocessing, Natural Language Processing based on spaCy, rule-based pattern matching and experimentation with ClinicalBERT.

The experimental pipeline was capable of recognizing several clinically relevant entities from the provided text, which included biomarker information (ER, PR, HER2, and Ki-67), as well as some selected tumor and lymph nodes information. Rule-based patterns proved particularly helpful when it came to extracting structured values often encountered in diagnosis reports.

The obtained information was transformed into structured fields and used as an input to other components of the proposed system. Thus, the basic feasibility of transforming unstructured clinical text into patient profile was shown.

Still, the quality of the extraction was highly dependent on the format of the input text. Differences in the clinical language, incomplete sentences, various problems with text formatting and ambiguity of some expressions impacted the consistency of the extraction. This indicated the necessity of further improving the process of entity extraction and testing it on the clinically annotated data set.

B. ClinicalBERT Experiment

The research was conducted with regard to exploring the application of ClinicalBERT language model as a part of the implemented experiment pipeline for processing medical text. In particular, it was needed to understand if contextual language understanding may help in extracting and interpreting clinical information.

It became clear that ClinicalBERT may be useful for providing context-based representation of the clinical text. However, in this experiment, the application of ClinicalBERT had several limitations associated with the existing input format, the number of samples in the experimental dataset, as well as with the lack of task-based training for this language model.

In other words, the experiment findings do not allow using ClinicalBERT as a decision-making tool separately. Rather, it is necessary to combine a rule-based pattern recognition and NLP with ClinicalBERT as a language model for processing complex texts.

C. Rule Extraction and Decision Engine Observations

A rule extraction technique was tested experimentally in order to explore whether treatment-related knowledge could be formalized into rules. The output included the conditions of the rule based on clinical features, an outcome of the treatment described by the rule, its confidence level, and an explanation.

As a result of the experiment, it has been observed that automatic rule extraction from clinical or guideline text can be noisy. Some rules extracted had conditions, outcomes, or explanations that had to be cleaned up and interpreted from the clinical point of view. Thus, it has become clear that automatically extracted rules should not be applied in clinical practice without verification.

Taking into account the conclusions made, the new system adopts a different technique for building the final decision engine. In particular, rules for the treatments are supposed to be explicitly formalized from already existing clinical guidelines and then added to the recommendation engine.

D. Discussion

The results of the initial experiments give an implementation-level proof-of-concept of the proposed workflow. It is proved that clinical data can be obtained from texts and represented in structured form for further processes of sub-type classification and reasoning about treatment based on rules.

At the same time, there were several challenges found during the experiments. A generic NLP pipeline cannot ensure successful extraction from any clinical report since hospitals may have different templates, terminologies, abbreviations, and report styles. Also, the language models like ClinicalBERT need corresponding data and task-oriented evaluation to be used reliably in the clinical extraction task.

It is confirmed by these results that the suggested hybrid approach is reasonable. Rule-based extraction gives transparency for clearly structured clinical expressions, whereas NLP and clinical language models can help to process more complex texts. The guideline-based decision engine introduces a completely different reasoning stage at which suggestions are made related to certain clinical conditions rather than predicted without explanation.

Also significant is the concept of the clinician-in-the-loop system. In the case of this system, the objective is to lower the burden associated with obtaining clinical information as well as verification of decision criteria. However, the final decision concerning the treatment is made by the physician.

E. Evaluation of the Complete System

Performance of the entire system will be analyzed on several levels. The NLP module will be analyzed using such metrics as precision, recall, and F1-score with regard to extraction of important clinical entities. The rule-based decision engine will be evaluated via comparison of produced treatment plans against the conditions of guidelines that apply to the cases.

Clinical validation might involve analysis by the medical experts regarding how relevant, accurate and explainable are the generated recommendations. Clinician feedback functionality may be used to mark whether the recommendation was accepted, changed or rejected.

However, a more extensive testing of the entire system should be performed with respect to breast cancer reports that are clinically validated and properly annotated.

VI. Conclusion and Future Work

In this paper, we propose an Explainable Rule-Based Clinical Decision Support System for Breast Cancer Treatment Support via Natural Language Processing. The proposed system is aimed at addressing the problem of gathering clinical information from different diagnostic reports and using this information for treatment support.

The proposed workflow includes clinical information extraction, data validation, molecular subtype classification, guideline-based treatment reasoning, safety check, and clinician's feedback. The key information such as ER, PR, HER2, Ki-67, tumor size, lymph node status, and disease stage are gathered and structured in the form of a patient profile. After that, the patient profile is used for the evaluation via explainable clinical rules for generation of potential treatments.

As a result of preliminary implementation experiments, it was proven that it is possible to extract clinical information related to breast cancer from the medical text via NLP methods, rule-based pattern recognition, and clinical language model building. During this process, some problems related to inconsistency in report format, ambiguity of clinical terminology, noise in the text, and automatically built rules were discovered. This observation inspired the proposal of a hybrid approach where structured extraction and clinically validated rules become a basis for decision support.

One of the features of the suggested approach is the fact that it supports, rather than replaces the doctor. The physician is able to make a decision regarding approval, changes or disapproval of the suggested treatment pathway. It is planned to include all the required patient information, related rules, as well as safety information in each recommendation in order to provide explainability.

Further research would involve completion and testing of the whole system using annotated reports of patients with breast cancer, which have been validated in a clinic. The NLP part can be improved by task-specific training and evaluation of the model in different types of report texts. The guideline-based rule base can also be improved and updated constantly using clinically approved recommendations.

In addition to that, further improvement can include a more comprehensive safety module that takes into consideration patient comorbidities, allergies, state of organs, drug interactions, contraindications to certain treatment options. Further research would also involve evaluation of the feedback mechanism using actual treatment decisions and results.

The system could be extended in the future by incorporating clinical data into medical imaging, histopathology and genomics through multimodal integration. This could help to establish a precision medicine platform, while at the same time preserving the clarity and rule-based decision-making process characteristic of the current system.

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